

Overview of Foundation model AI

Subject: Foundation model AI

1.

General-purpose AI Due to their adaptability to a wide range of use-cases, foundation models are sometimes considered to be examples of general-purpose AI. In designing the EU AI Act, the European Parliament has stated that a new wave of general-purpose AI technologies shapes the overall AI ecosystem. The fuller structure of the ecosystem, in addition to the properties of specific general-purpose AI systems, influences the design of AI policy and research. General-purpose AI systems also often appear in people's everyday lives through applications and tools like ChatGPT or DALL-E. Government agencies like EU Parliament have identified regulation of general-purpose AI, such as foundation models, to be a high priority. General-purpose AI systems are often characterized by large size, opacity, and potential for emergence, all of which can create unintended harms. Such systems also heavily influence downstream applications, which further exacerbates the need for regulation. In regards to prominent legislation, a number of stakeholders have pushed for the EU AI Act to include restrictions on general-purpose AI systems, all of which would also apply to foundation models.

2.

Supply chain Foundation models' general capabilities allow them to fulfill a unique role in the AI ecosystem, fueled by many upstream and downstream technologies. Training a foundation model requires several resources (e.g. data, compute, labor, hardware, code), with foundation models often involving immense amounts of data and compute (also referred to as computational power). Due to foundation models' large development costs and inexpensive adaptation requirements, the AI landscape has shifted to a small subset of AI companies making foundation models for downstream adaptation. Thus, most foundation model companies outsource this step to specialized data providers (e.g. Scale AI, Surge) and compute providers (e.g. Amazon Bedrock, Google Cloud, Microsoft Azure).

3.

History Quanta Magazine traced world models back to a 1943 publication by Kenneth Craik on mental models and the blocks world of SHRDLU in the 1960s. Business Insider traced world models to a 1971 paper by Jay Wright Forrester. A related idea of organizing world knowledge, the frame representation, was proposed by Marvin Minsky in 1974. In 2018, researchers David Ha and Jürgen Schmidhuber defined world models in the context of reinforcement learning: an agent with a variational autoencoder model V for representing visual observations, a recurrent neural network model M for representing memory, and a linear model C for making decisions. They suggested that agents trained on world models in environments that simulate reality could be applied to real world settings. In 2022, Yann LeCun saw a world model (defined by him as a neural network that acts as a mental model for aspects of the world that are seen as relevant) as part of a larger system of cognitive architecture – other neural networks that are analogous to different regions of the brain. In his view, this framework could lead to commonsense reasoning. LeCun has estimated that world models would be fully functional by the late 2020s to mid 2030s.

4.

History Technologically, foundation models are built using established machine learning techniques like deep neural networks, transfer learning, and self-supervised learning. Foundation models differ from previous techniques as they are general purpose models that function as a reusable infrastructure, instead of bespoke and one-off task-specific models. Advances in computer parallelism (e.g., CUDA GPUs) and new developments in neural network architecture (e.g., Transformers), and the increased use of training data with minimal supervision all contributed to the rise of foundation models. Foundation models began to materialize as the latest wave of deep learning models in the late 2010s. Relative to most prior work on deep learning, these language models demonstrated the potential of training on much larger web-sourced datasets using self-supervised objectives (e.g. predicting the next word in a large corpus of text). These approaches, which draw upon earlier works like word2vec and GloVe, deviated from prior supervised approaches that required annotated data (e.g. crowd-sourced labels). The 2022 releases of Stable Diffusion and ChatGPT (initially powered by the GPT-3.5 model) led to foundation models and generative AI entering widespread public discourse. Further, releases of LLaMA, Llama 2, and Mistral in 2023 contributed to a greater emphasis placed on how foundation models are released with open foundation models garnering a lot of support and scrutiny.

5.

Release strategies After a foundation model is built, it can be released in one of many ways. There are many facets to a release: the asset itself, who has access, how access changes over time, and the conditions on use. All these factors contribute to how a foundation model will affect downstream applications. In particular, the two most common forms of foundation model release are through APIs and direct model downloads. When a model is released via an API, users can query the model and receive responses, but cannot directly access the model itself. Comparatively, the model could be directly downloadable for users to access and modify. Both release strategies are often classified as an open release. The exact definition of an open release is disputed, but widely accepted requirements are provided by the Open Source Initiative. Some open foundation models are: PaLM 2, Llama 2, Granite, and Mistral. While open foundation models can further research and development more easily, they are also more susceptible to misuse. Open foundation models can be downloaded by anyone, and particularly powerful models can be fine-tuned to intentionally or unintentionally cause harm. During a closed release, the foundation model cannot be accessed by the public, but is used internally by an organization. Such releases are considered safer, but offer no additional value to the research community or the public at large. Some foundation models like Google DeepMind's Flamingo are fully closed, meaning they are available only to the model developer; others, such as OpenAI's GPT-4, are limited access, available to the public but only as a black box; and still others, such as Meta's Llama 2 are open, with broadly available model weights enabling downstream modification and scrutiny.

6.

Training World models are trained on a variety of data modalities, including text, images, audio and video, and have been applied to video generation. One open source dataset for world models includes 1 billion data points across multiple modalities (text, images, audio, video and point clouds), including 1 million manual annotations.